

Technology sustainability assessment of biodiesel development in South Africa: A system dynamics approach

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ABSTRACT

Currently, several renewable energy technologies have reached, or are approaching maturity and can be utilised for small- and large-scale applications. Biodiesel, a cleaner burning replacement fuel, is argued to potentially contribute to sustainable development in many countries and regions of the world. Biodiesel development, however, entails complex interactions of actors such as the technology developers, government at different levels, communities, as well as the natural environment; different actions or responses in the greater system might hinder or undermine the positive effects of such a development. Given such complexity, an integrated and holistic technology assessment approach is indispensable to determine the potential effects of biodiesel development on sustainability in general, which can then inform and enhance proper planning and management. The paper thus introduces the Bioenergy Technology Sustainability Assessment (*BIOTSA*) model that has been developed, based on a system dynamics approach. The model is demonstrated and evaluated with a specific case: the assessment of the effects of biodiesel development on selected sustainability indicators for the Eastern Cape Province of South Africa. The strengths and limitations of the model are discussed and future perspectives are outlined.

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1. Introduction

Many studies are recognizing the potential of renewable and alternative energy to contribute to sustainable development [1,2]. However, the development of renewable and cleaner energy technologies still involves interactions with the environment. For instance, most renewable energy options require land for their development, but may have the potential to reduce emissions. Renewable energy also has a social function in human lives and interactions may be established between the energy development and applicable social systems.

Biodiesel is one of the renewable and alternative energy technologies that is argued to contribute to sustainable development at global, national and community levels [2]. Biodiesel development, however, involves activities that affect the environment, economy

and society. In addition, biodiesel production development involves diverse actors including, among others, the policy makers; technology developers/investors; technology assessment practitioners; and the community that would be involved in growing crops for the biodiesel production. These factors display the characteristics of a complex system that constitutes biodiesel production development, which Sukkasi et al. [2] argue might undermine the achievement of sustainable development. Thus, biodiesel technology management and careful planning is imperative.

One of the important disciplines in technology management is technology assessment, which has evolved over the past four decades [3]. While technology assessment does not necessarily provide policy makers and managers ‘the answer’, it does increase the odds that the maximum benefits of technology will be achieved [4] especially concerning sustainable development. Some studies in the literature acknowledge the need to evaluate the renewable and clean energy technology development as a complex system [5–7]. Thus, the approach to use in assessing the renewable and alternative energy technology development for sustainability will need to be in a position to account for the assumptions regarding sustainable development [8,9]. This includes the economic, social,

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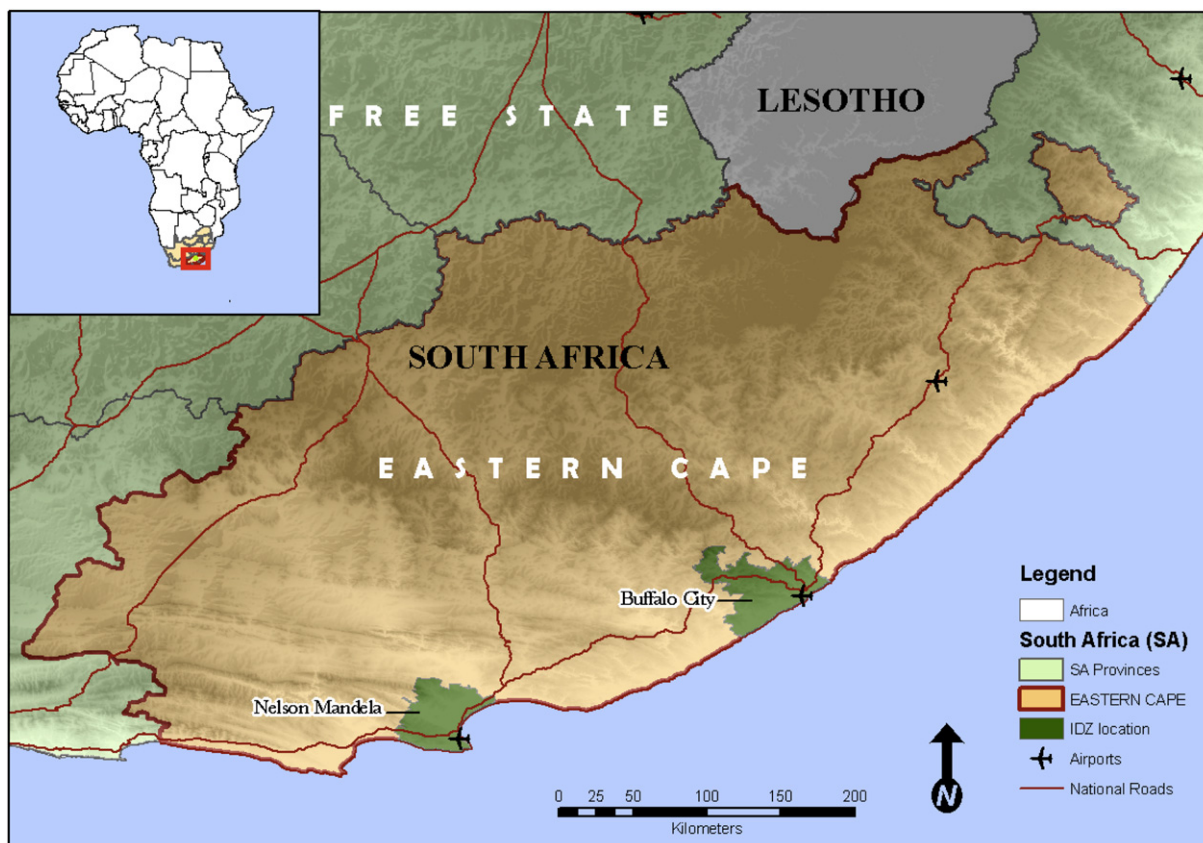


Fig. 1. Map showing Eastern Cape Province, South Africa.

environmental and other changes that might influence its development towards the desired sustainable path. System dynamics provides a potential for such an assessment [6,10,11].

System dynamics is among the tools and methods that are recognized for technology assessment [3,4]. It is an interdisciplinary and transdisciplinary approach that is based on the theory of system structures [12]. System dynamics represents complex systems and analyses their dynamic behaviour over time [13]. The approach has gained popularity due to its focus on the structure and its flexibility, and its potential as an intermediate level tool in technology assessment is recognized [14].

Recently, several studies have used system dynamics to analyse renewable and alternative energy related issues. Tesch et al. [15] developed a system dynamics model of global agricultural and biomass development. Flynn and Ford [16] modelled and simulated carbon cycling and electricity generation from energy crops. Bantz and Deaton [17] used system dynamics to envision possible growth scenarios for the United States biodiesel industry over a course of a decade. Scheffran et al. [18] developed a spatial-dynamics model of energy crop introduction in the state of Illinois. Pruyt and De Sitter [19] developed a highly-aggregated system dynamics model of the worldwide food/bioenergy to investigate interactional effects between agricultural food production and bioenergy production worldwide. In the South African context, Musango et al. [20] developed an energy model as an initial step of building a Threshold 21 (T21) model for South Africa. Their study, however, did not focus on the development of biofuel value chains. Musango et al. [21] present the bioenergy systems sustainability assessment and management (BIOSSAM) framework, which is the first system dynamics approach in South Africa focussing on biodiesel development.

Although bioenergy system dynamics models have already been developed for several regions and for bioenergy related aspects, this does not seem to be the case for all regions and all aspects [19]. The use of system dynamics in studies related to bioenergy technology policies and their effects on sustainability at local, national and regional levels is therefore imperative. The purpose of this study is thus to present the Bioenergy Technology Sustainability Assessment (BIOTSA) model which was developed to examine the effect of biodiesel development interventions on selected sustainability indicators for the Eastern Cape Province of South Africa.

2. Biodiesel development in the Eastern Cape Province

The development of the biodiesel industry in the Eastern Cape Province of South Africa (see Fig. 1) is part of a new agrarian transformation plan for the Province and one of the priority projects under the Accelerated and Shared Growth Initiative for South Africa (ASGI-SA). ASGI-SA is a South African government strategy intended to boost economic growth and reduce unemployment in the country.

While no commercial biodiesel production yet exists, a number of these projects are in the pipeline and are planned for the Eastern Cape Province. One of these biodiesel projects, which is used as a case in this paper, plans to establish a plant that utilizes canola as a feedstock and produce about 400 million metric tons of biodiesel [23]. The project aims to avoid the usual criticism of food insecurity associated with biofuels, as canola is a winter crop that would be cultivated in rotation with maize or wheat, which are summer crops. They plan to use the agrarian transformation model that is presented in Fig. 2. The proposed biodiesel plant is envisaged to

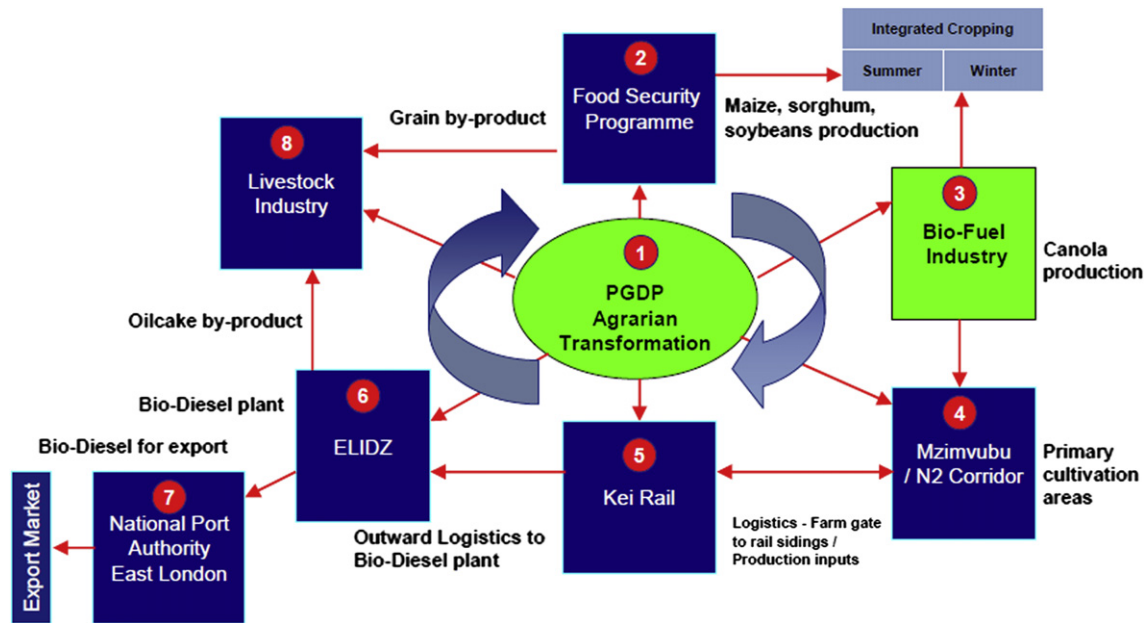


Fig. 2. Proposed agrarian transformation model [23].

stimulate a large market for agricultural products, including canola, soybeans and sunflower, which were not in the Province [22].

The planned biofuel refinery for this project is likely to be in the Coega Industrial Development Zone (IDZ) in the Nelson Mandela Municipality of the Eastern Cape Province (see Fig. 1). IDZ's are purpose-built industrial estates geared for duty-free production for exports, and they play a significant role in South Africa's macro-economic policy. They provide transport, logistics and business services tailored for export-oriented industries. The infrastructure development is first class as they are next to airports and seaports, and have direct road and rail links [24]. Finally the biodiesel produced from this project is destined for the European market, where "cold proof" fuel, which can be used in winter in Europe, is required.

3. Bioenergy technology sustainability (BIOTSA) model development

The general methodological framework for energy technology sustainability assessment is discussed elsewhere [11]. This section provides a discussion of how the framework was utilized in the development of the BIOTSA model.

3.1. Identifying biodiesel technology development needs in the Eastern Cape Province

The initial process of the BIOTSA model development was the identification of the need for developing biodiesel production and the existing features of biodiesel production development in the Eastern Cape Province. This was based on secondary data sources, information/data obtained through the desktop search, personal communication with the researchers at the Department of Agriculture in Bisho and a survey visit in the Eastern Cape Province in South Africa. The Department of Agriculture in Bisho is involved in some trial experiments for crops for biofuel production, and the communication with some of the key researchers facilitated the understanding of the state of the development of biodiesel crops production in the province. On the other hand, site visits to some of the potential areas earmarked for biodiesel crop production in the

Eastern Cape Province took place during June 2009. A more detailed discussion of the survey results is found in Amigun et al. [28].

In the Eastern Cape Province, the identified needs for developing biodiesel production are outlined in the South Africa biofuels strategy [29]. These needs are mainly to address the issues of rural poverty, rural development and Black Economic Empowerment.

3.2. Identifying the sustainability indicators for biodiesel development in the Eastern Cape Province

The BIOTSA model was meant to assess the sustainable development of biodiesel production in the Eastern Cape Province. It was thus important to identify the sustainability indicators related to such a development. The major interactions considered between society, economy and environmental systems of biodiesel production development are presented in Fig. 3: (i) the society component included population, community perception and employment, which were relevant for the social system in the Eastern Cape Province; (ii) the economic component, which invests capital and labour for biodiesel production in the Eastern Cape economy; and (iii) the environmental components, which determined the key resources use, such as land, water, energy and biodiesel related wastes.

Information and data on the economic, environmental and social conditions in the Eastern Cape, as presented in Fig. 3, were gathered from various sources. Data on population, employment, Eastern Cape investments (economy), land use and water demand and supply was obtained from: (i) Statistics South Africa [25]; and (ii) a 2005 development report by the United Nations Development Program, the Development Bank Southern Africa, and the Human Resources Scientific Council. Data on biofuels capital investment, biodiesel production, and potential biodiesel crop land were obtained from various websites such as Engineering News [26]; bio-energy site [27]; Department of Minerals and Energy documents, currently the Department of Energy; personal communications with the Industrial Development Zone in East London, South Africa; and a survey with the community in the Eastern Cape communities by means of person to person semi-structured interviews and a focus group discussion [28].

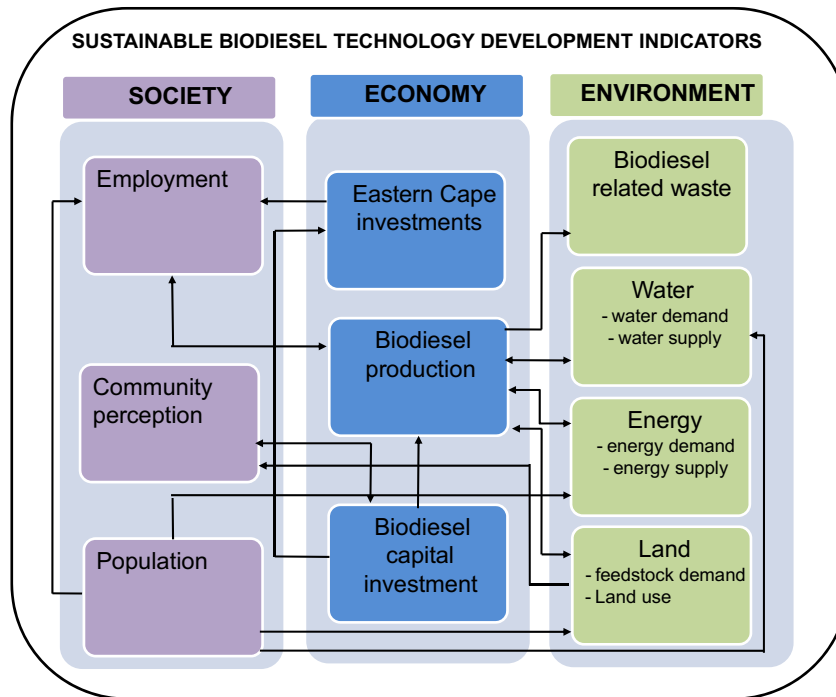


Fig. 3. Society-economy-environment interactions in biodiesel production development Source: Adapted from [21].

This information and data were necessary to populate the environmental, economic and social impact/indicators related to biodiesel production development. Sustainability indicators involve either direct quantitative measurement and/or qualitative assessments given quantitative rankings. Criteria for identifying and selecting adequate sustainability indicators are proposed by several authors. According to Hardi and Zdan [30], the selection of indicators should be based on policy relevance, simplicity, validity, availability of time-series data, good quality, affordable data, the ability to aggregate information, sensitivity to small changes and reliability. Another issue in developing indicators is the number of indicators, typically exemplified in the so called information pyramid. While these considerations should be taken into account, some authors warn of the difficulty in finding indicators that comply with a subgroup of the criteria [31].

It is important to note that some indicators are directly related to a specific technology under consideration. Evans et al. [32] and Evans et al. [33] provide some sustainability indicators which were used as a basis for renewable energy technology assessment. Other indicators are dependent on the economic, environmental, social and political context in which the technology is being implemented. Thus, additional sustainability indicators for the biodiesel development in the Eastern Cape were identified during interactions with some identified stakeholders.

Data limitations are often a significant obstacle to generating large indicator sets, and this study was not an exception to such a limitation. The final choice of indicators that were used in this study was thus driven by four main factors: (i) indicators that had most relevance to the study subject; (ii) indicators that reflected the Eastern Cape Provincial issues of biodiesel production development; (iii) indicators encompassing each of the environment, social and economic and other spheres of sustainability; and (iv) the ability to quantify the indicators and data availability.

Based on the above criteria, ten indicators were identified and selected for the biodiesel technology assessment and are presented in Table 1. The set includes three economic indicators, namely biodiesel production, biodiesel profitability and Eastern Cape GDP

per capita; two social indicators, namely employment created due to biodiesel plant development and the community acceptance to grow crops for biodiesel production; and five environmental indicators, namely air emissions resulting from biodiesel production, land use changes due to the introduction of the biodiesel production, water and energy use in the biodiesel production, and biodiesel by-products particularly glycerol.

3.3. System dynamics modelling

Once the needs and sustainability indicators for biodiesel development in the Eastern Cape Province were identified, the next step was to develop a *BIOTSA* model following the system dynamics theory. The *BIOTSA* model was divided into eleven sub-models. The *BIOTSA* sub-models represent the society, economy and environment interactions of biodiesel production (see Fig. 3). These sub-models are: biodiesel production, cost of operation, biodiesel production profitability, GDP, employment from a biodiesel plant, community perception, population, land, water, emissions and electricity demand. Each sub-model and selected equations for variables are described in the subsequent sections.

3.3.1. Biodiesel production sub-model

The biodiesel production sub-model is one of the core systems of the *BIOTSA* model. Presently in South Africa, there is no commercial biodiesel production [34] and only plans are in place to construct the plant. The biodiesel project that was used as a case for this paper is based on proposed biodiesel plant that will be operational in 2012. The structure of the biodiesel production is partly taken from the generic commodity market model [12] and adapted to suit the conditions of the proposed biodiesel production development in the Eastern Cape. The structure that represents the biodiesel production is demonstrated in Fig. 4.

This sub-model consists of three key stocks; these are, biodiesel capacity construction, functional biodiesel capacity and the accumulated biodiesel. The biodiesel capacity construction (*BC*, in litre) is increased by the rate in which biodiesel plant construction starts

Table 1
Sustainability indicators for bioenergy technology assessment in Eastern Cape.

	Indicator	Symbol	Description	Units
Economic	Biodiesel production	ECO ₁	This measures the quantity of biodiesel production	Litre/year
	Biodiesel profitability	ECO ₂	This measures the profitability from biodiesel production	USD/year
	Eastern Cape GDP	ECO ₃	This measures the per capita GDP in the Eastern Cape Province	USD/person/year
Social	Employment	SOC ₁	This measures the labour force participation due to the investment in the biodiesel plant capacity	Person
	Community perception	SOC ₂	This is represented by the effect of community perception on land conversion for biodiesel production crops and measures the community acceptance to grow these crops	Dimensionless
Environmental	Land use change	ENV ₁	This measure the changes in land use due to the introduction of biodiesel production. This includes changes in fallow land, agricultural land, biodiesel crop land and livestock land.	Ha
	Air emission	ENV ₂	This measures the total avoided air emissions due to investment in biodiesel production	kg CO ₂ /year
	Biodiesel by-product	ENV ₃	This measures the amount of accumulated glycerol resulting from biodiesel production.	Litre/year
	Water use	ENV ₄	This measures water use as a result of biodiesel production	Litre/year
	Energy use	ENV ₅	This measures energy use as a result of biodiesel production	KWh/year

(r_{BC} , in litre/year) and decreased by new biodiesel capacity (r_{NBC} , in litre/year) once the construction is completed. The dynamics of biodiesel capacity construction is thus mathematically represented as:

$$BC(t) = BC(0) + \int [r_{BC} - r_{NBC}] dt \quad (1)$$

Where $BC(0)$ is the initial value of the biodiesel capacity construction, which is zero. The new biodiesel capacity is determined by the biodiesel capacity construction (BC , in litre) divided by the biofuel construction time (t_{BC} , in year).

$$r_{NBC} = \frac{BC}{t_{BC}} \quad (2)$$

On the other hand, biodiesel plant construction start (r_{BC} , in litre/year) is determined by future biodiesel capacity investment ($FBCI$, in USD/year) divided by the capital cost per year (CC_a , in USD/litre). Thus:

$$r_{BC} = \frac{FBCI}{CC_a} \quad (3)$$

The future biodiesel capacity investment ($FBCI$, in USD/year) is the sum of the endogenously determined desired biodiesel capacity (DBC , in litre/year) and exogenous planned biodiesel investment (PBI , in litre/year), multiplied by the capital cost per year (CC_a , in USD/litre). Thus:

$$FBCI = [DBC + PBI] * CC_a \quad (4)$$

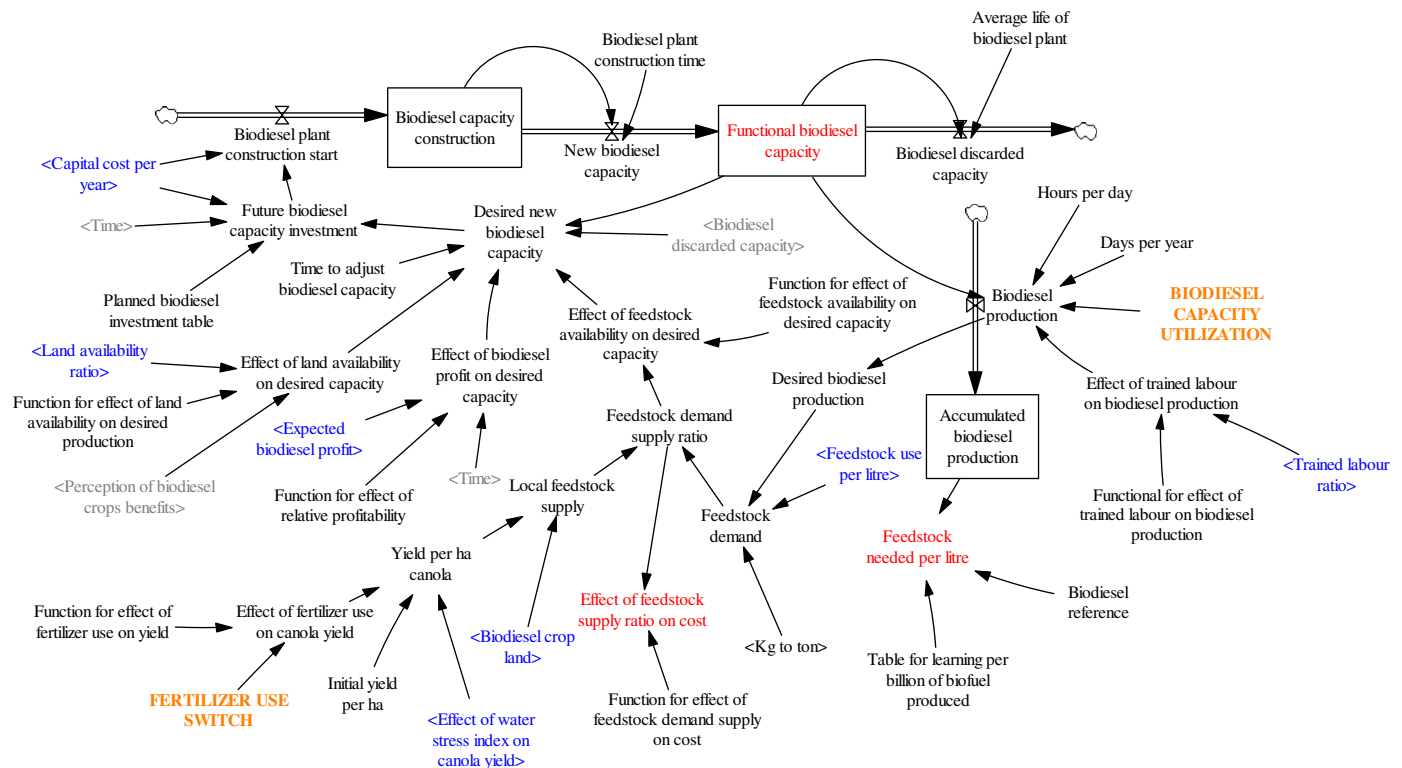


Fig. 4. The stock and flow diagram of the biodiesel production sub-model of the BIOTSA model.

Desired biodiesel capacity (DBC , in litre/year) is an important variable in the biodiesel production sub-model and it is mathematically represented as:

$$DBC = \left(\frac{FBC}{t_{ABC}} + BDC \right) * E_{BPC} * E_{FA} * E_{LA} \quad (5)$$

Where FBC is functional biodiesel capacity (in litre); t_{ABC} is the time to adjust biodiesel capacity (in year); BDC is the biodiesel discarded capacity (litre/year); E_{BPC} is the effect of biodiesel profit on desired capacity (in dimensionless units); E_{FA} is the effect of feedstock availability on the desired capacity (in dimensionless units); and E_{LA} is the effect of land availability on desired production (in dimensionless units).

The effect of biodiesel profit on desired capacity was used as a non-linear function of the expected profit. Such a non-linear function was made possible by using the VENSIM's^{®1} lookup table. Lookup tables are generally tables that stores numerical data in either simple two dimensional or multi-dimensional arrays. Lookup tables are used in system dynamics modelling when a modeller is faced with a situation where a relationship exists between two variables yet a simple algebraic equation that defines the relationship is non-existence. Thus, table functions are preferable to complicated equations since the modeller can have control on the slopes, shapes and saturation points to accurately provide a representation of two variables in a non-linear relationship. In addition, lookup tables can facilitate clear understanding of the model and facilitate participation from non-specialist. Lookup tables are generated by experiments or creating artificial data of input and output of the system. In this study, no experiments were carried out and thus, the lookup were generated artificially based on the expected theoretical behaviour and discussion with some key persons in the field. All the input variables were normalized and the units for both inputs and outputs were dimensionless units.

The biodiesel plant that is in operation in the *BIOTSA* model is regarded as functional biodiesel capacity (FBC , in litre), and it is increased by the new biodiesel capacity (r_{NBC} , in litre/year) and decreased by the biodiesel discard capacity (r_{BDC} , in litre/year). This is represented as:

$$FBC(t) = FBC(0) + \int [r_{NBC} - r_{BDC}] dt \quad (6)$$

The equation for the biodiesel discarded capacity (r_{BDC} , in litre/year) is represented as the functional biodiesel capacity (FBC , in litre) divided by the average life of biodiesel plant (t_{ALBP} , in year)

$$r_{BDC} = \frac{FBC}{t_{ALBP}} \quad (7)$$

Given the functional capacity of the biodiesel plant, the biodiesel production can be estimated. This is calculated by multiplying the functional biodiesel capacity multiplied with the duration of operation of the plant in a year. In addition, biodiesel production is not only dependent on the functional capacity but also on the labour availability. The assumption is that the labour at the biodiesel plant level comes from the Eastern Cape Province. Since biodiesel production is new to the Eastern Cape, the trained labour is fundamental in influencing the amount of biodiesel production. The *BIOTSA* model initially assumes 100% capacity utilization. However in practice, firms will not produce at 100% utilization as this allows no scope for fluctuation. Effective capacity

utilization of about 90% has however been recorded elsewhere [35] but for Africa, effective capacity utilization could be lower due to the local specific situation [36]. In addition, the *BIOTSA* model assumes that the biodiesel plant will be operational for 330 days in order to account for shut down due to maintenance purposes. Thus, biodiesel production (r_{BP} , in litre/year) is given as:

$$r_{BP} = FBC \left(\frac{24 \text{ hour}}{\text{day}} * \frac{330 \text{ day}}{\text{year}} \right) * E_{TL} * CU \quad (8)$$

Where: E_{TL} is the effect of trained labour on biodiesel production; and CU is the biodiesel capacity utilization (in dimensionless units).

The effect of trained labour on biodiesel production is used as a non-linear function following the concept of diminishing returns of the standard production function [37,38].

While biodiesel production (r_{BP} , in litre/year) is determined by the functional biodiesel capacity, it is also an inflow to the accumulated biodiesel production (ABP , in litre) which is the third stock of the biodiesel production sub-models. This is an important stock in biodiesel production sub-model because it is used to estimate the learning curves that result from accumulated production. This is represented as:

$$ABP(t) = ABP(0) + \int [r_{BP}] dt \quad (9)$$

3.3.2. Land sub-model

Land use changes are one of the increasing concerns in the biodiesel production development [39]. This is because of other human activities particularly food production requiring land use besides growing energy crops. Thus, land availability is critical issue in many countries [39,40].

Eastern Cape is the second largest Province in South Africa, with a land size of 168,966 square kilometres. The land use in the Eastern Cape Province is divided into six competing land use categories namely: settlement land (S , in ha), forest plantations (F_p , in ha), fallow land (F , in ha), food crop land (C_f , in ha), conservation land (C , in ha), livestock land (L , in ha). An expected seventh land use in the Eastern Cape is the biodiesel crop land (C_b , in ha), which is as a result of biodiesel production development. The land sub-model thus included the seven different types of land uses in the Eastern Cape Province. The structure representing the land sub-model is presented in Fig. 5.

Population dynamics influences settlement, food crop and livestock land requirements. The South African government plans to establish biodiesel crop land from fallow land. This makes fallow land one of the key land stocks because it can also be converted to other forms of land, depending on the land requirements. Only the dynamics of the fallow land and biodiesel crop land are discussed here. Thus, the stock of fallow land is given as:

$$FL(t) = FL(0) + \int \left[\sum (r_{LF}, r_{C_fF}, r_{C_bF}) - \sum (r_{FC_f}, r_{FL}, r_{FC_b}, r_{FC}, r_{FF_p}, r_{FS}) \right] dt \quad (10)$$

where: r_{LF} is the rate of conversion from livestock land to fallow land (in ha/year); r_{FC_f} is the rate of conversion from fallow land to food crop land (in ha/year); r_{BC_f} is the rate of conversion from biodiesel crop land to fallow land (in ha/year); r_{FL} is the rate of conversion from fallow land to livestock land (in ha/year); r_{FC} is the rate of conversion from fallow land to biodiesel crop land (in ha/year); r_{FC} is the rate of conversion from fallow land to conservation land (in ha/year); r_{FF_p} is the rate of conversion from fallow land to forest plantations (in ha/year); and r_{FS} is the rate of conversion from fallow land to settlement land (in ha/year).

¹ <http://www.ventana.com>

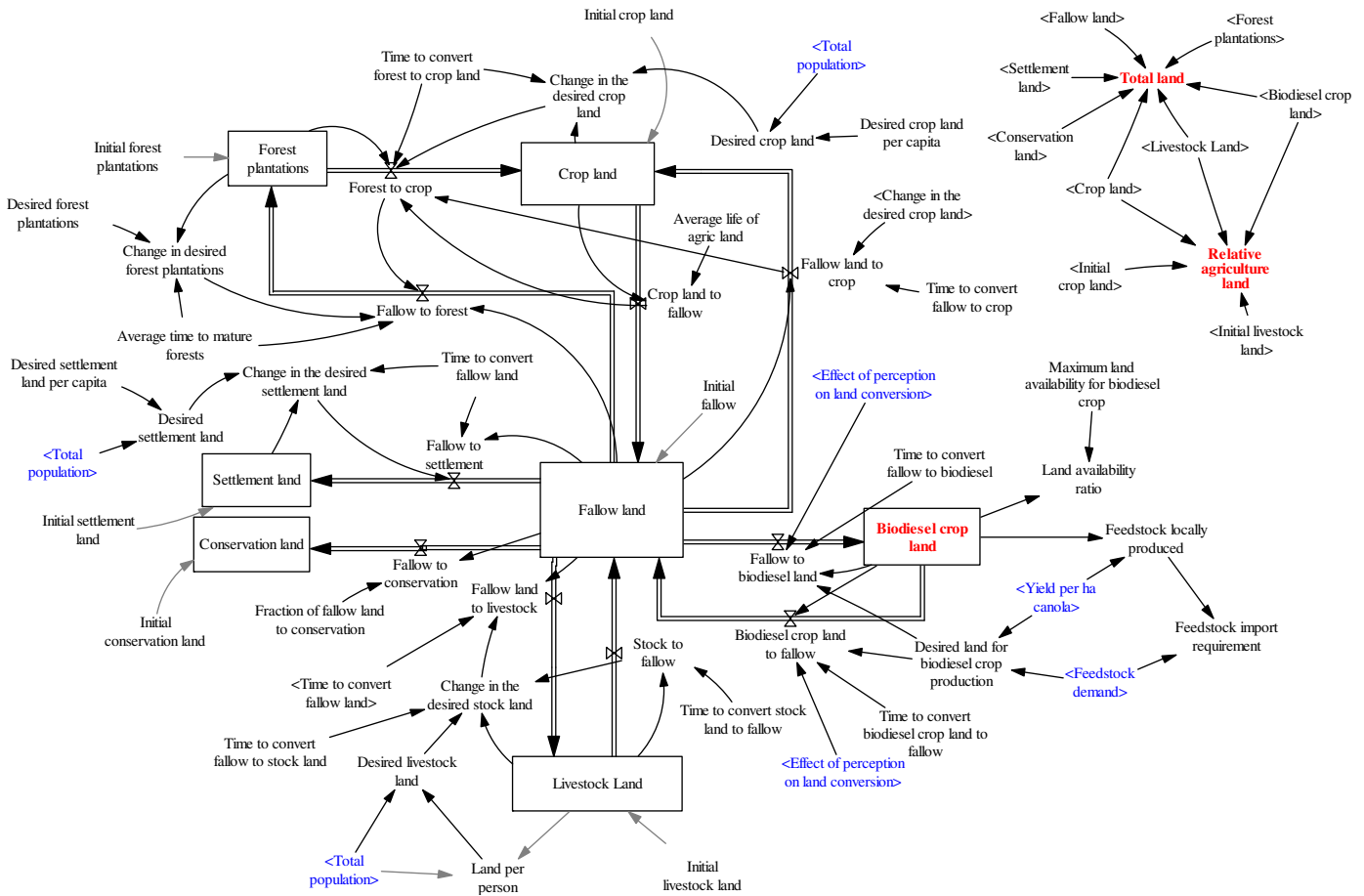


Fig. 5. The stock and flow diagram of land sub-model of the BIOTSA model.

The stock of biodiesel crop land on the other hand is mathematically represented as:

$$BCL(t) = BCL(0) + \int [r_{FC_b} - r_{BC_f}] dt \quad (11)$$

$$r_{FC_b} = \text{Max} \left[0, \left(\frac{DBCL - BCL}{t_{FC_b}} \right) * E_{PBL} \right] \quad (12)$$

Where, $DBCL$ is the desired biodiesel crop land (in ha); BCL is biodiesel crop land (in ha); and E_{PBL} is the effect of perception on land conversion (in year).

3.3.3. Biodiesel profitability sub-model

Profitability of any production activity is important in determining the future investments.

Some of the most important factors determining the biodiesel plant profitability are the cost of production particularly the feedstock costs, biodiesel production and the biodiesel plant capacity [17,41]. This sub-model thus represents the profitability of the biodiesel production and consists of one stock, the expected biodiesel profit; and one flow rate, that is, changes in expected biodiesel profit. The expected biodiesel profit represents the balance between price and cost and is expressed by a dimensionless number. The expected biodiesel profit is an important driver of the desired new biodiesel capacity. Fig. 6 shows the stock and flow diagrams of the expected biodiesel profit (EBP , in dimensionless units), driven by the changes in expected biodiesel profit (r_{CBP} , in dimensionless units/year). This is mathematically represented as:

$$EBP(t) = EBP(0) + \int [r_{CBP}] dt \quad (13)$$

$$r_{CBP} = \left(\frac{UBP - EBP}{t_{AP}} \right) \quad (14)$$

Where UBP is unit biodiesel profitability (in dimensionless units) and t_{AP} is the time it takes to adjust profit (in year).

The unit biodiesel profitability is determined by the unit total cost of production (UTC , in USD/litre) and the biodiesel price (P_b , in USD/litre). Hence:

$$UBP = \left(\frac{P_b - UTC}{P_b} \right) \quad (15)$$

The biodiesel unit total cost of production is the sum of costs such as: electricity cost, water use cost, feedstock cost, labour cost, maintenance cost and other administration costs. The support for biodiesel is subtracted from these costs. The potential support of R0.53 (USD 0.07)² per litre for biodiesel production as stated in the South Africa biofuel Draft Strategy [29] is not accounted for in the initial analysis in order to test the effect of this policy. Following Amigun et al. [36], this study used the selling price of the conventional fuel as proxy for biodiesel price. This is because there is no current data that exists for biodiesel price. A switch was also used for the unit total cost of a production in order to test a scenario where the biodiesel plant sells its by-production such as glycerol

² The exchange rate used was 1USD = 7.5 Rand.

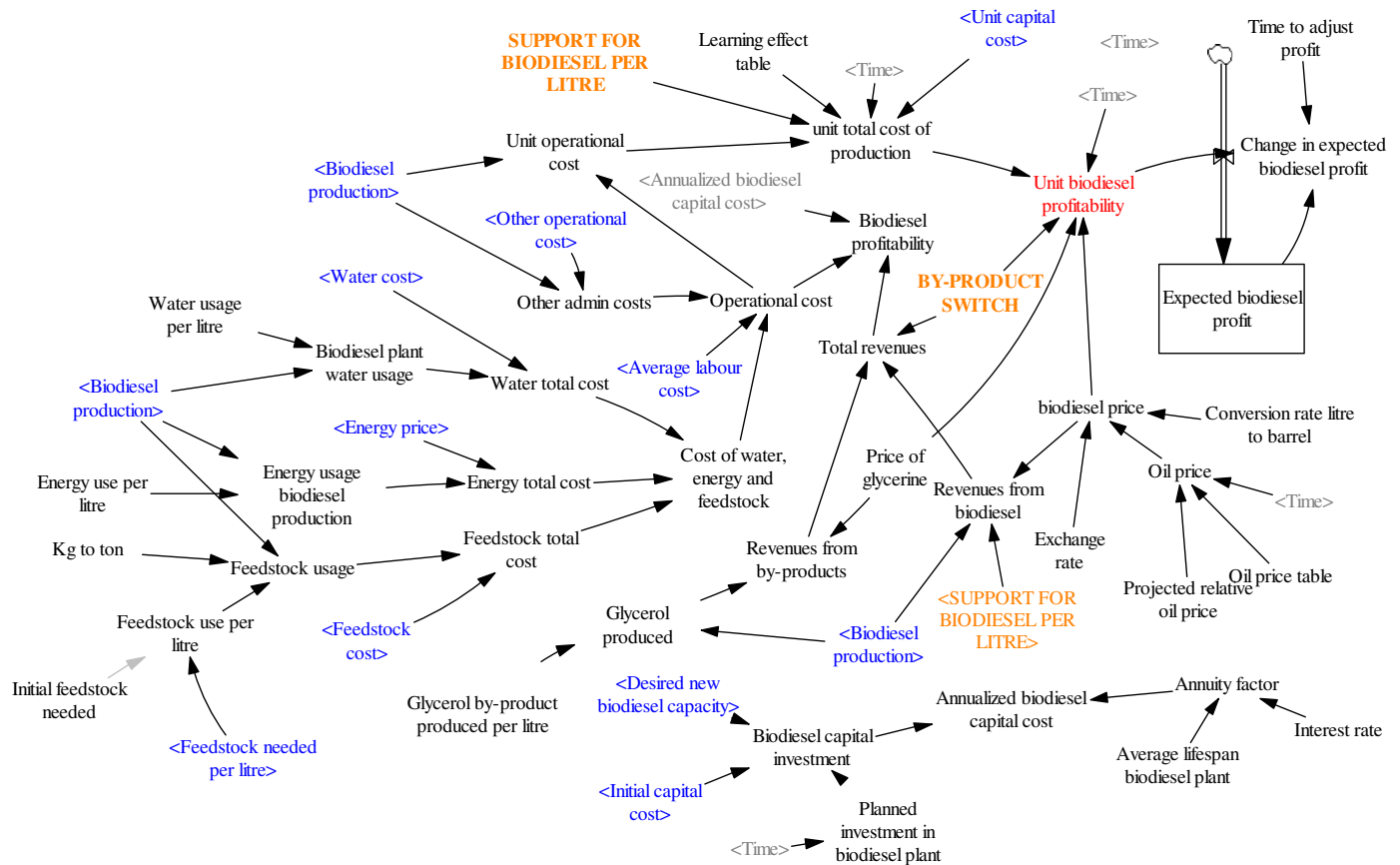


Fig. 6. The stock and flow diagram of biodiesel profitability sub-model of *BIOTSA* model.

and cake. A switch takes a value of 1 if the glycerol is part of revenue generation and zero if not.

3.3.4. Cost of production sub-model

This sub-model is important since high cost of biodiesel development is recognized as a major barrier to large-scale commercialization [42]. There are a number of costs that are involved in the biodiesel production. This includes capital costs and operational costs such as feedstock, water cost, energy cost and other operational costs. Feedstock cost is the dominant cost in biodiesel production [36]. This sub-model represents such costs and consists of four stocks: feedstock cost, water cost, other operational costs and unit capital cost, which are influenced by an exogenous fractional rate. All the four costs have more or less similar structure and are presented in Fig. 7. To avoid repetition in the discussion, only the dynamics of the feedstock costs are explained.

Feedstock cost (FDC , in USD/ton) is influenced by the changes in the feedstock price (r_{FP} , in USD/ton/year) and it is represented as:

$$FDC(t) = FDC(0) + \int [r_{FP}] dt \quad (16)$$

On the other hand, the changes in feedstock price are determined by both feedstock cost growth which is a constant and feedstock demand-supply behaviour. The feedstock supply-demand dynamics is based on the notion that, if the feedstock is not produced locally, the demand for feedstock is met from imports which has a cost implication.

3.3.5. Employment from biodiesel plant sub-model

The employment from biodiesel plant is explored to determine the number of jobs that are created as a result of establishing the

biodiesel plant in the Eastern Cape. Employment is important in improving the economy of the Eastern Cape Province. Thus, the employment sub-model (Fig. 8) is designed to calculate the desired employment from biodiesel plant ($DEBP$, in person) and the process of recruiting the desired employment in order to achieve a trained workforce that matches with the desired employment. There is two stock variables, workforce in training (WiT , in person) and employment biodiesel plant (EBP , in person); and two flow rates, recruiting for training (r_{rt} , in person/year) and new trainees (r_{NT} , in person/year). The desired employment in the biodiesel plant is limited to the plant size and it makes use of an equation developed by Amigun et al. [36]. Thus:

$$DEBP = \alpha(FBC)^\beta \quad (17)$$

Where: α is the capacity coefficient; and β is the employment elasticity.

The information on the desired employment biodiesel plant is fed into the recruiting for training (r_{rt} , in person/year), which determines the dynamics of the workforce in Training (WiT, in person). In addition, employment biodiesel plant (EBP , in person) and net recruitment noise (NRN , in person/year) determines the recruitment for training. These relationships are represented as:

$$WiT(t) = WiT(0) + \int [r_{rt}]dt \quad (18)$$

On the other hand, the recruiting for training is given as:

$$r_{rt} = \left(\frac{DEBP - EB}{t_{AT}} \right) + NRN[\text{RANDOMUNIFORM}(-0.5, 0.5, 0)] \quad (19)$$

Where t_{AT} is the time to adjust training (in year).

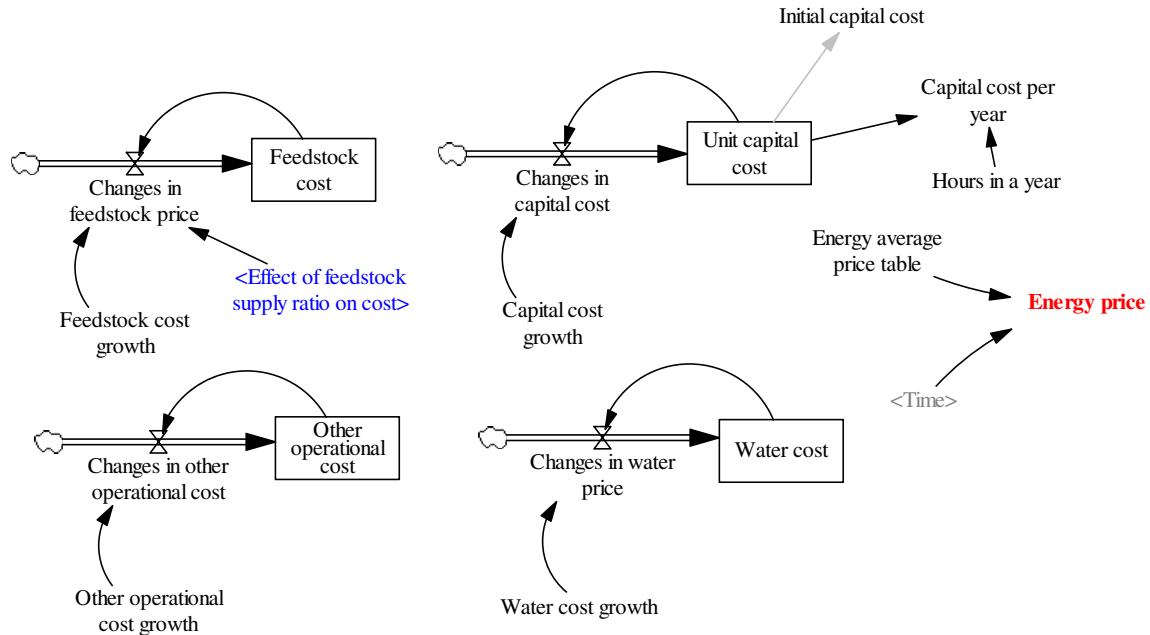


Fig. 7. The stock and flow diagram of cost of operation sub-model of BIOTSA model.

The net recruitment noise (NRN , in person/year) in the Equation (19) follows a random uniform, with a minimum value of -0.5 and maximum value of 0.5 , which was assumed for this analysis. The employment biodiesel plant on the other hand is represented as:

$$EBP(t) = EBP(0) + \int [r_{NT}]dt \quad (20)$$

$$r_{NT} = \left(\frac{WiT}{t_{NT}} \right) \quad (21)$$

Where r_{NT} are new trainees (in person/year); and t_{NT} is time to complete training (in year).

3.3.6. Water sub-model

Water is one of the resources required for both biodiesel plant and at the biodiesel crop production level. These two types of water requirements form part of the water demand. Water consumption is however significant at the biodiesel crop production stage and can rely on irrigation or rainfall. The production of the biodiesel crop in the Eastern Cape Province is expected to rely on rainfall. The water

sub-model (Fig. 9) thus aims to calculate the water stress index (WSI , in dimensionless units) in the Eastern Cape Province, which influences the production yield and thus determining the land requirements for biodiesel crop production.

$$WSI = \left(\frac{TWD}{TRWR} \right) \quad (22)$$

Where TWD the total water is demand (in kg/year); and $TRWR$ is the total renewable water resources (in kg/year).

Total water demand is the sum of domestic and municipal water demand ($DMWD$, in kg/year) and the production water demand (PWD , in kg/year) represented as:

$$TWD = \sum (DMWD, PWD) \quad (23)$$

3.3.7. Electricity demand sub-model

Energy, particularly electricity is an important input in the biodiesel production process and thus necessary to represent its contribution to the existing electricity demand. The electricity

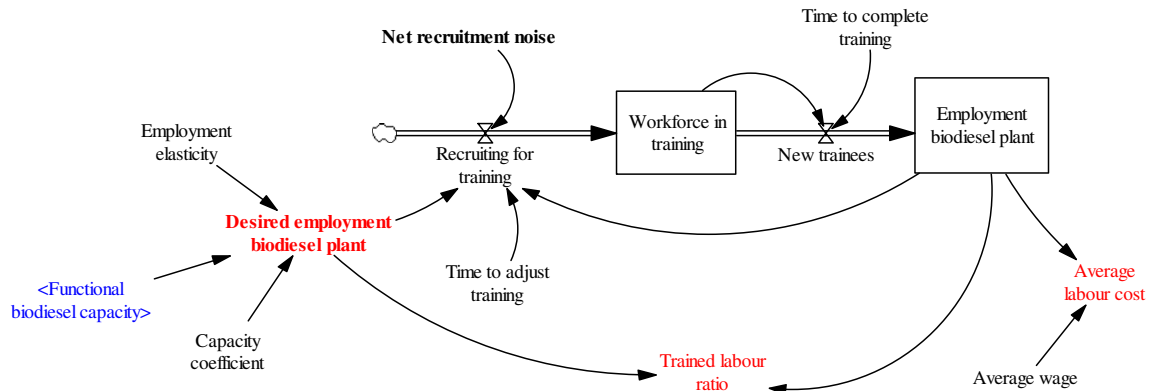


Fig. 8. The stock and flow diagram of employment biodiesel plant sub-model of BIOTSA model.

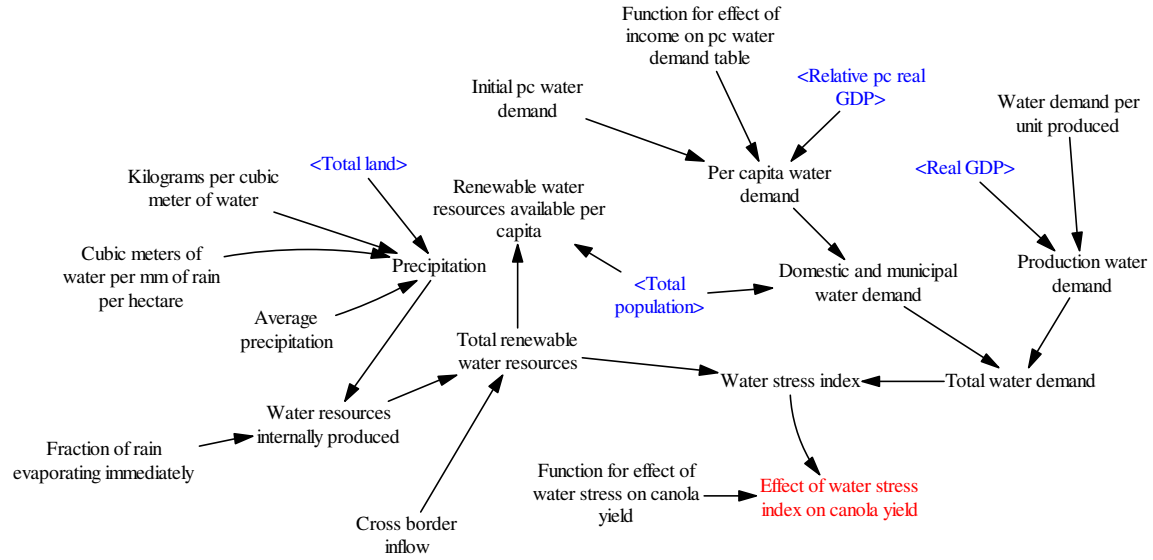


Fig. 9. The stock and flow diagram of water sub-model of BIOTSA model.

demand sub-model (Fig. 10) thus shows the electricity demand in the Eastern Cape which is driven by the economic activities, including the biodiesel production and population.

The dynamics of the electricity demand (ED , in MWh) is determined by the changes the electricity demand (r_{ED} , in MWh/year). These relationships are represented as:

$$ED = ED(0) + \int [r_{ED}] dt \quad (24)$$

$$r_{ED} = \left(\frac{[ED(RP)^\varphi * (RRGDP)^\mu] - ED}{t_{AE}} \right) + CEU \quad (25)$$

Where RP is the relative population (in dimensionless units); φ is the elasticity of population to demand (in dimensionless units); $RRGDP$ is the relative real GDP (in dimensionless units); μ is elasticity of GDP to demand (in dimensionless units); t_{AE} is the time to

adjust electricity demand (in year); and CEU is changes in energy usage (in MWh/year).

3.3.8. Emissions sub-model

While biodiesel is a cleaner technology, there are emissions that are associated with its production. Biodiesel emissions are mainly dependant on the feedstock used and the biodiesel production model. Generally, these emissions in carbon dioxide equivalent, is relatively lower than its diesel counterpart. The emissions sub-model (Fig. 11) thus calculates the total emissions and the emissions avoided due to biodiesel production which in turn determines the net emissions (NE) from the biodiesel production.

The sub-model consist of one stock variable, cumulative net emissions, and two flow rates, air emission generation and air emission decomposition. Thus, the stock of cumulative net emissions (CNE , in kg CO₂) is increased by air emission generation (r_{EG} ,

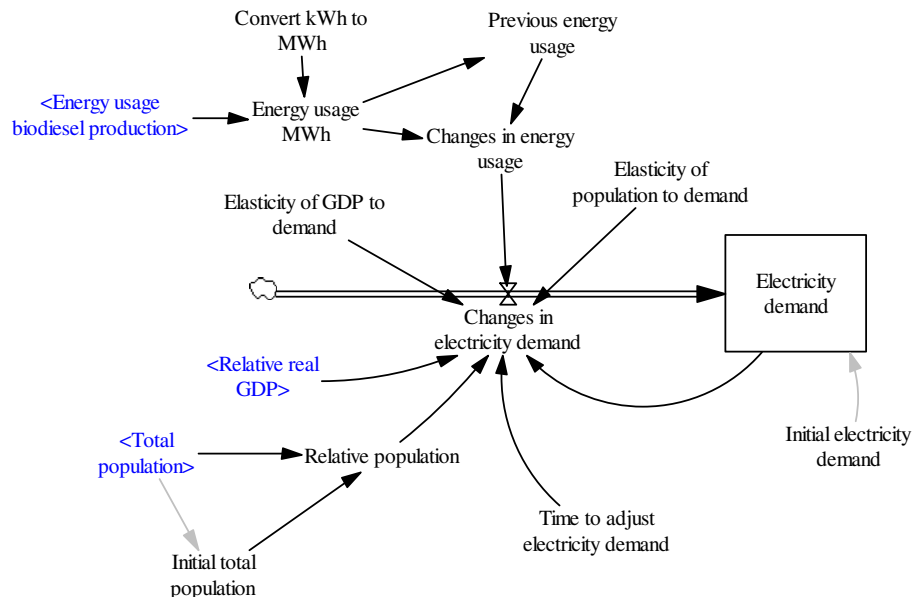


Fig. 10. The stock and flow diagram of the electricity demand sub-model.

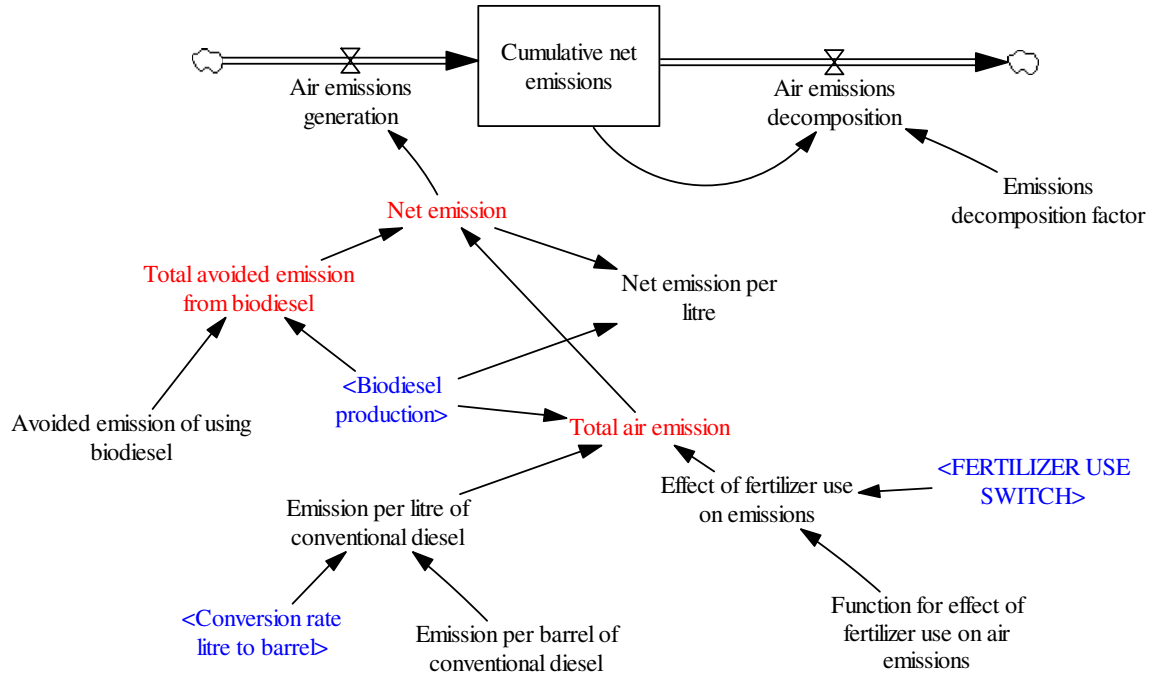


Fig. 11. The stock and flow diagram of the air emissions sub-model.

in kg CO₂/year) and decreased by the air emissions decomposition (r_{ED} , in kg CO₂/year). Thus,

$$CNE = CNE(0) + \int [r_{EG} - r_{ED}]dt \quad (26)$$

Air emission generation (r_{EG} , in kg CO₂/year) is determined by the net emissions (NE , in kg CO₂/year) and is given as:

$$r_{EG} = NE \quad (27)$$

On the other hand, the net air emissions (NE , in kg CO₂/year) is the difference between the total air emissions (TAE , in kg CO₂/year) and the total avoided emission from biodiesel ($TAEB$, in kg CO₂/year). Thus:

$$NE = TAE - TAEB \quad (28)$$

The total air emissions (TAE , in kg CO₂/year) include emissions that would be generated if the conventional diesel was produced and the resultant emissions if fertilizer is used in the biodiesel crop production. A switch was used to test fertilizer use and it takes a value of 1 if the fertilizer is used in biodiesel crop production (canola) and zero if not. The total air emissions is thus mathematically represented as avoided emissions per litre of conventional diesel (ECD , in kg CO₂/litre) multiplied by a biodiesel production (BP , in litres/year) and the effect of fertilizer use on emissions (EFU , in dimensionless units).

$$TAE = ECD * BP * EFU \quad (29)$$

The total avoided emission from biodiesel ($TAEB$, in kg CO₂/year) is given as avoided emissions of using biodiesel (AEB , in kg CO₂/litre) multiplied by a biodiesel production (BP , in litre/year).

$$TAEB = AEB * BP \quad (30)$$

3.3.9. Population sub-model

Population is an important sub-model and it is used to determine some social and economic indicators. The structure of

the population sub-model (Fig. 12) is customized from Millennium Institute Threshold-21 [43] to fit within the scope of this study. The population in the BIOTSA model was defined as the Eastern Cape Province population, which was categorized according to sex and age groups. The key stock variable is population (P , in person), which was defined by three rate variables namely: births (r_B), deaths (r_D , in person/year) and net migration (r_M , in person/year). Population stock dynamics was therefore given as:

$$P = P(0) + \int [r_B - r_D - r_M]dt \quad (31)$$

Births and death rates are both influenced by the economic conditions which in turn influence the level of the population stock. Changes in population do influence the water demand, electricity demand, and GDP.

3.3.10. GDP sub-model

One of the main factors of promoting biofuel investments in the Eastern Cape Province is to improve its economy. This is because the Eastern Cape Province contribution to the GDP is among the least in South Africa. An explicit representation of the Eastern Cape GDP was therefore crucial.

This sub-model consists of two stocks as shown in Fig. 13. The first one is the South Africa GDP which is driven by an exogenous GDP growth. The importance of this stock is to estimate the total investment in the Eastern Cape (TI_{EC} , in USD/year). The second stock is, capital stock (K , in USD). Both total investment in the Eastern Cape (TI_{EC} , in USD/year) and future biodiesel investments (FBI , in USD/year) influence the gross capital formation (r_{GKF} , in USD/year), which in turn influences the capital stock. The capital stock (K , in USD) is however decreased by the rate of depreciation (r_{DN} , in USD/year).

$$K = K(0) + \int [r_{GKF} - r_{DN}]dt \quad (32)$$

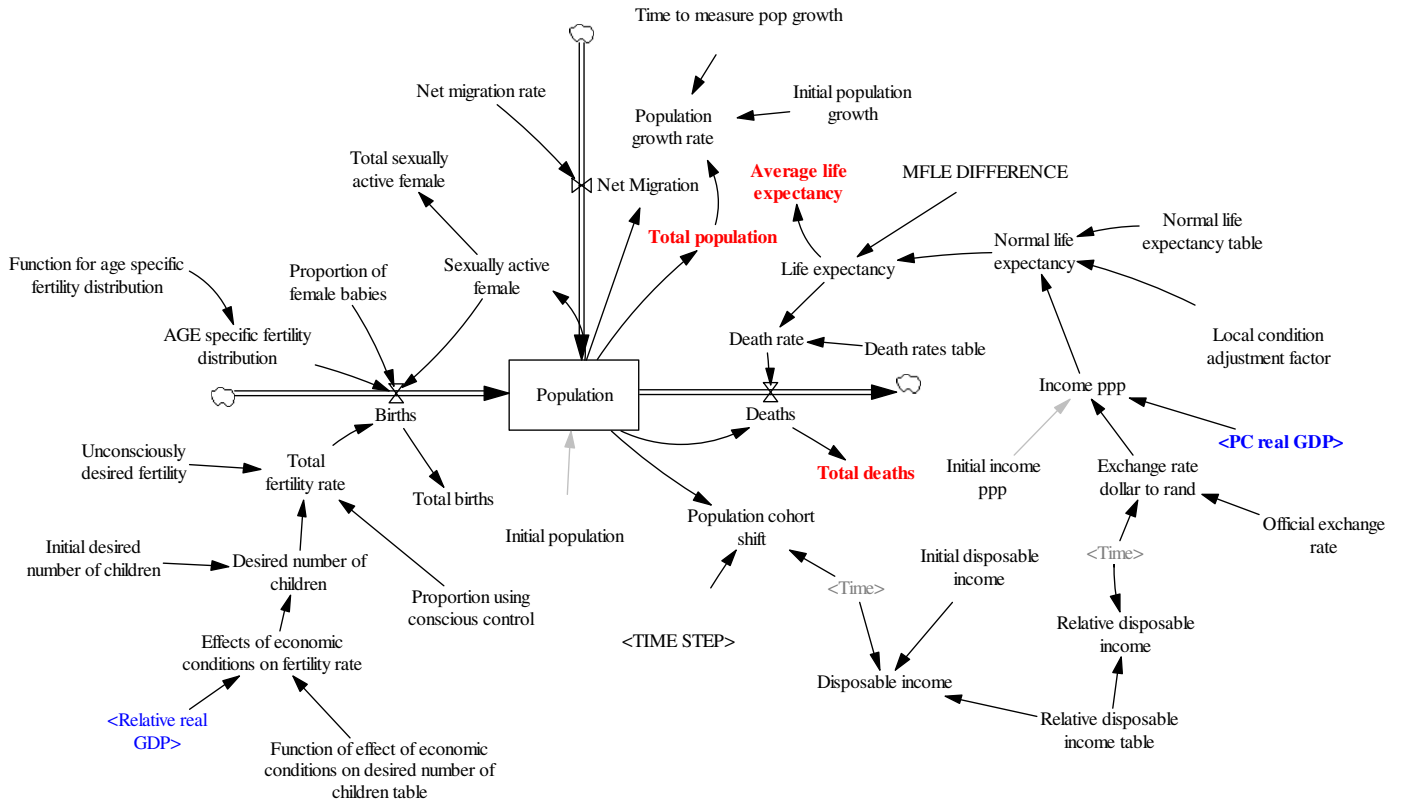


Fig. 12. The stock and flow diagram of population sub-model of BIOTSA model.

Based on the Cobb–Douglas production function [44], capital (K , in USD) and labour (L , in person) are the key factors determining GDP. Thus:

$$RGDP = RGDP(0) * RP \quad (33)$$

Where $RGDP$ is the real GDP (in USD/year); $RGDP(0)$ is initial real GDP (in USD/year); and RP is relative production (dimensionless units).

On the other hand, relative production (RP , in dimensionless units) is calculated as:

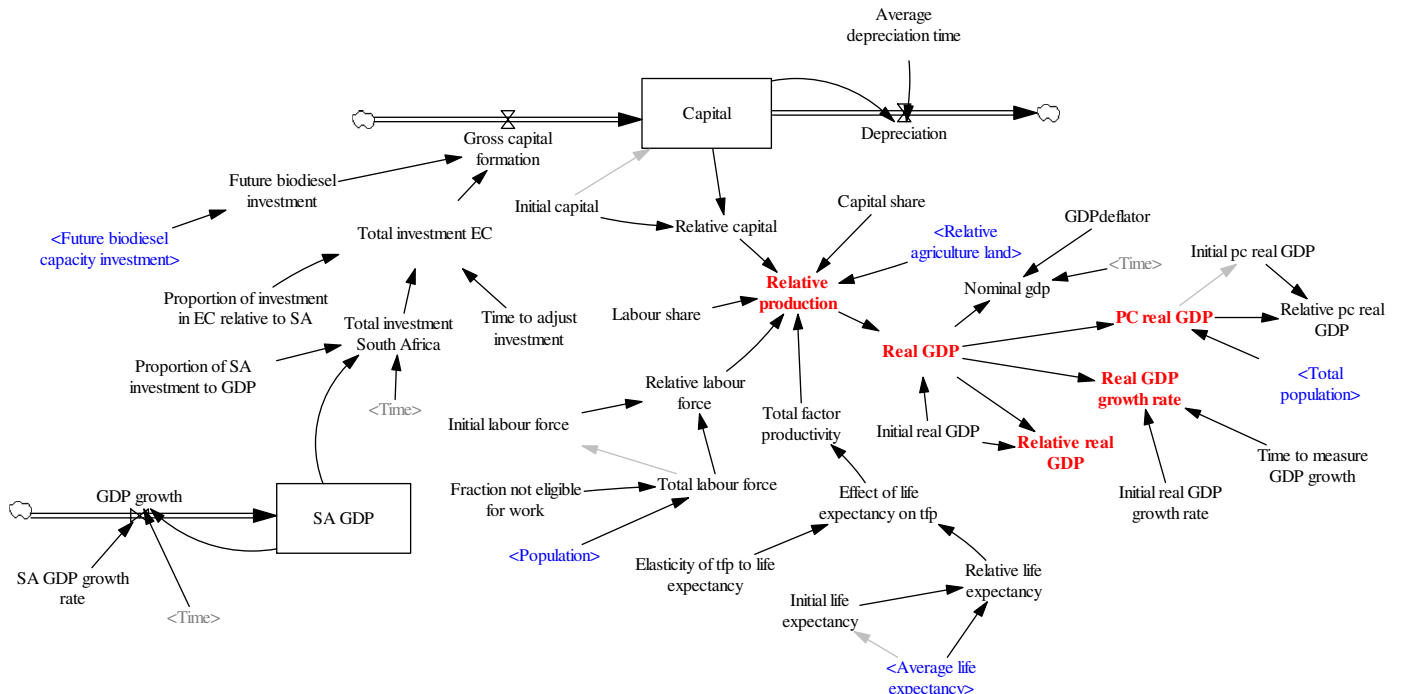


Fig. 13. The stock and flow diagram of the GDP sub-model of BIOTSA model.

the model [49]. According to Barlas [49] *structural validity tests* determine how well the structure of the model matches the structure of reality. This is the case when every model component has a real world counterpart and when every key factor contributing to the problem in the real world has a model counterpart. As descriptions of system structure are generally not available, they have to be extracted from the mental models of people familiar with the system. In this study, the structural validation process began at the initial stage of model building when the first few parameters were estimated and their relationships defined. This continued throughout the entire model building and each of this is discussed.

4.1.1. Direct structure test

The direct structure validity test was carried out to check the models consistency with the knowledge of the real system relevant to the purpose. A number of approaches were applied to test the structural validity tests. Firstly, the *BIOTSA* model utilised case specific data of Eastern Cape Province in South African and/or knowledge available about the real system of biodiesel development. Secondly, the mathematical relationships for the *BIOTSA* model were evaluated and comparisons made with the real situation of the biodiesel production development. As an illustration, Table 2 shows three equations of how this process was carried out for all the equations in the model.

In addition to the mathematical equations, tests were done to ensure that the *BIOTSA* model conforms to the basic physical conservation laws. An example of a common violation of this law entails a stock that becomes negative. For instance, all land stocks can either be zero or a positive number, but not a negative value. Thus the outflow tests from these flows should approach zero as any of these stocks approaches zero. The first equation in Table 2 provides such an illustration. This was inspected for all the stocks and their respective flows.

Thirdly, some existing structure of some sub-models were adopted and thus served as a “theoretical” direct structure test [47] for *BIOTSA*. The conclusion that was drawn from undertaking all these tests was that the *BIOTSA* model is a robust simplification of the processes occurring in the real world.

4.1.2. Dimensional consistency test

This test seeks to verify that the equations in the model are dimensionally consistent without using arbitrary constants that do not represent the real world situation [12]. The test was done by inspecting the dimension of inputs for the equations and also from the actual simulation of the model. The simulation using the VENSIM® software did not generate any dimension consistency error.

4.1.3. Parameter confirmation test

This test was performed throughout the model-building phase. Since biodiesel production in South Africa is yet non-existent, an

attempt was made to ensure that all the parameters defined represent the real system of biodiesel development in South Africa. Some control buttons were included for some parameters which were in question so that they could be specified with a particular range. In doing so, this allowed for prompt and effective extreme condition testing and scenario analyses. The standard statistical test was not possible to be applied in this study because of lack of historical data. Apart from the data limitation, the test could not be possible even with the availability of data due to the transient, non-stationary behaviour of most of the model indicators. According to [48], the modelled problem is of no statistical nature and validating it statistically is unsuitable due to multicollinearity and autocorrelation problems. The appropriate approach suggested by Barlas [48] is the use of graphical/visual measures to compare the behaviour pattern.

4.1.4. Extreme condition test

A model that has the ability to properly function when subjected to extreme conditions contributes to its utility as a policy evaluation tool and increases user confidence. Two types of extreme condition tests are recognized in system dynamics namely: structure-oriented and direct behaviour test. The former was used for the *BIOTSA* model in order to assess the plausibility of the resulting behaviour against the anticipation and/or knowledge of what is expected to occur under similar condition in the real world [50]. A number of selected parameters were thus assigned extreme values and a comparison made on the model-generated behaviour to the anticipated behaviour of the real system under a similar extreme condition. A selected output of some of the sustainability indicators is discussed below:

Test 1: Extreme initial community perception.

The initial community perception of biodiesel crops parameter was set to 1 and the results from such a simulation for the selected sustainability indicators affected by this parameter are presented in Fig. 15. In reality, if such a situation happens, it implies a high level of community acceptance to grow biodiesel production crops implying larger amount of biodiesel crop land as observed in Fig. 15a. With large biodiesel crop land available, there would be a higher incentive for the desired biodiesel capacity which obviously increases the and overall biodiesel production as shown in Fig. 15b. Improved profitability and total avoided emissions increased as a result of increased biodiesel production (Fig. 15c and d respectively). It is thus evident from the result in Fig. 15 that the simulation properly responds to the set extreme condition.

Test 2: Extreme feedstock cost growth rates.

The baseline results of *BIOTSA* model used the feedstock cost growth rates of 1% which were based on assumption from expert opinion. When such cost growth rates are increased to 25% the simulation results are shown in Fig. 16. With such a condition in the real world, higher increases in the feedstock cost decreases the

Table 2
Selected examples of direct structure test.

Equation	Explanation with real situation
$r_{FC_b} = \text{Max} \left[0, \left(\left(\frac{DBCLBCL}{t_{FC_b}} \right) * E_{PBL} \right) \right]$	Land that is converted for biodiesel crops comes from fallow land. The conversion of this land is however dependent of the community perception and acceptance of growing these crops and the desired biodiesel crop land by the investors of the biodiesel plant. The fuzzy maximum function (Sterman, 2000) is used because the conversion rate of fallow land to biodiesel crop land must remain non-negative.
$r_{BP} = FBC \left(\frac{24 \text{ hour} * 330 \text{ day}}{\text{day} \text{ year}} \right) * E_{TL}$	Biodiesel production is the output from a biodiesel plant and is only possible based on the capacity that is functional, the number of hours in a year it runs and the labour.
$DEBP = \alpha(FBC)^\beta$	Employment is the number of people working in the biodiesel plant and this is dependent on the capacity that is functional.

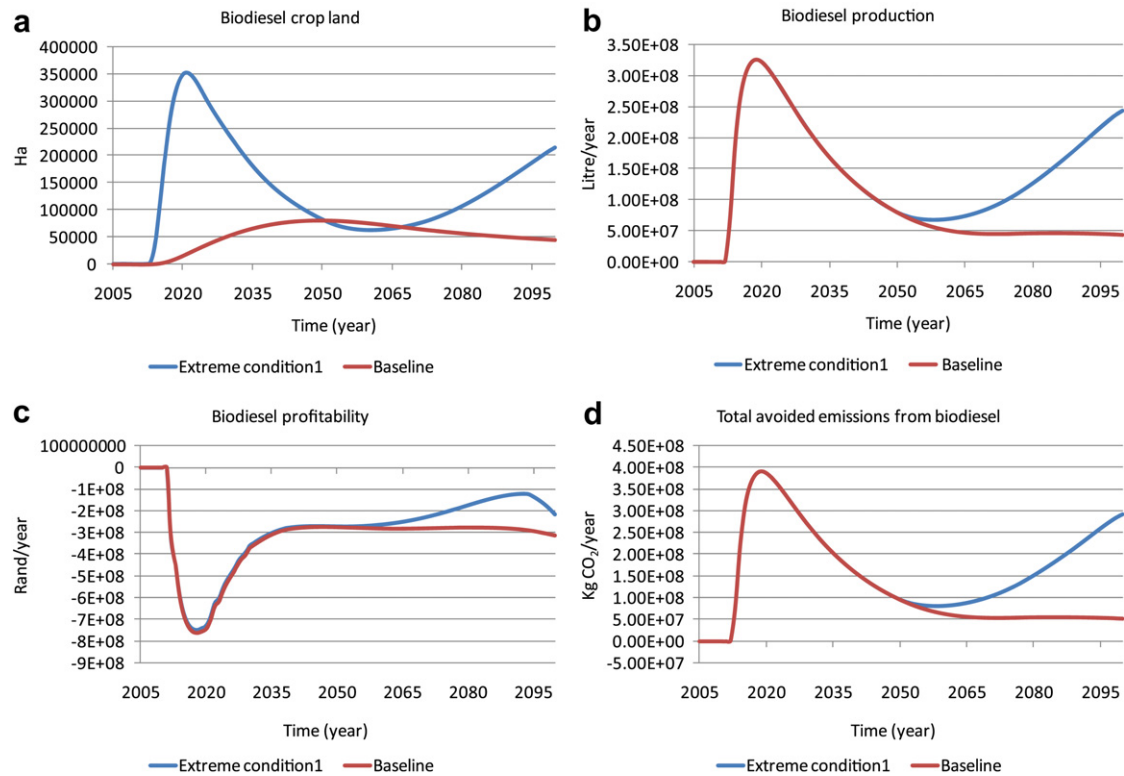


Fig. 15. Extreme condition 1 of initial community perception results.

profitability of biodiesel production (see Fig. 16a) which in turn reduces the desired biodiesel plant capacity to zero (see Fig. 16b). The reduction in the capacity implies reduced biodiesel production. Lower biodiesel production implies reduced revenue generation further decreasing the biodiesel profitability. It is also clearly seen that the simulation results in Fig. 16 responds to the extreme condition of increased costs growth rate.

Test 3: Planned biodiesel investment.

When the planned biodiesel investment is set to zero, it is expected that there is no future development of the biodiesel plant, which means no conversion of land to biodiesel crop land, no biodiesel production and thus implying zero profitability. The simulated results in Fig. 17 accurately respond to this condition.

Generally, the extreme condition results indicate that *BIOTSA* model does generate the right results for the right reasons.

4.2. Behavioural validity

Behaviour tests determine how consistently model outputs match real world behaviour [48]. This can either be based on available time-series data or the correlation of mental models with established reference models [12]. The usefulness of the former clearly depends on the quality of the available historical data, while the latter necessitates a substantive and coherent overlap in mental models. There are a number of ways of undertaking behavioural tests and how they were carried out for the *BIOTSA* model is discussed below.

4.2.1. Reference test

This entails running the simulation a few years or decades in the past and comparing how the model reproduces the key indicators with the actual, historical data. Unfortunately, this test was not possible with the *BIOTSA* model due to the non-existence of the

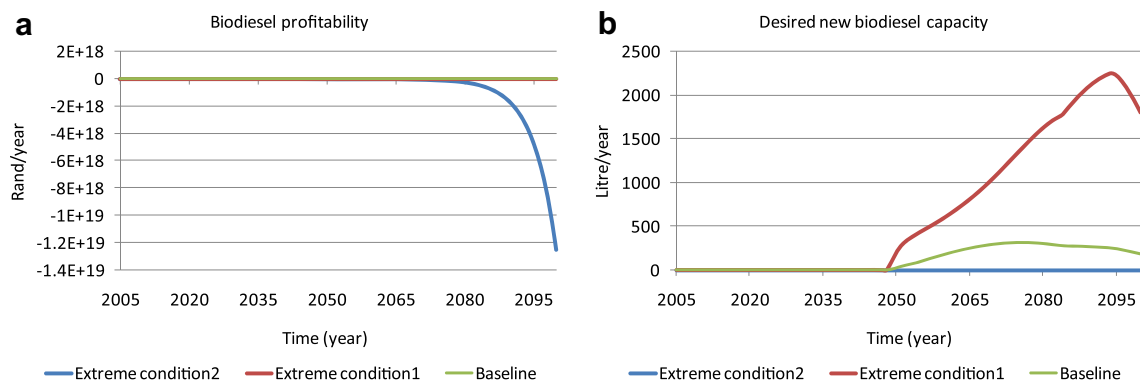


Fig. 16. Extreme condition 2 of cost growth rates result.

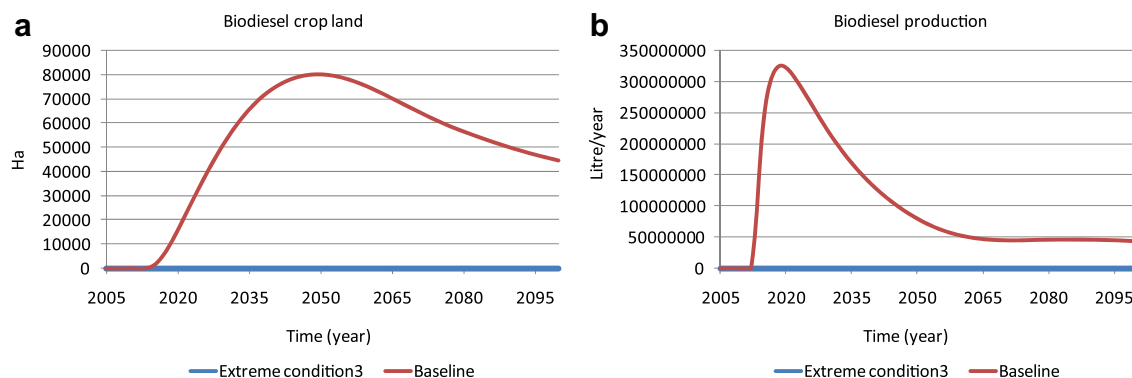


Fig. 17. Extreme condition 3 planned biodiesel investment table.

biodiesel development market in South Africa. Thus, no historical data exists that could be used to undertake the reference test.

4.2.2. Sensitivity analysis

It was important to know how the *BIOTSA* model is sensitive to different assumptions. The VENSIM® software does provide a platform for carrying out robust Monte-Carlo sensitivity simulations. Monte-Carlo simulation explores the uncertainty and future possibility of selected output variable through a selected number of repeated simulations, in which the uncertain or unknown parameters are randomly drawn. Given the uncertain parameters, confidence bounds are used for illustrating model validity. As part of the validation process for the *BIOTSA* model, the focus on the sensitivity analysis was on the cost growth rates with particular focus on the feedstock. This is because the price of feedstock such as canola seed is determined by the import parity and local supply, which vary not only from year to year, but also from month to month. Thus, biodiesel production and profitability would be greatly influenced by the cost of feedstock due to changes in price and it is necessary to establish how sensitive the production is as a result of the variation in price. Thus, the feedstock cost and other cost growth rates were set between -5% and 5% and the behaviour of the selected sustainability indicators were observed. In addition, it was uncertain on the length of time to change the community perception which was set between 1 year and 20 years. Table 3 provides the range of the parameters that were to change in order to carry out the sensitivity test. A random uniform distribution was used for all these parameters and the number of simulations was set at 400 scenarios.

Fig. 18 provides the simulation based confidence bounds for biodiesel crop land, biodiesel profitability, biodiesel production and employment in the biodiesel plant. These results indicate the percentage test cases in the Monte-Carlo simulation that fall within a particular percentage of confidence bounds. For instance, 100% of biodiesel productivity test cases are located in the 100% confidence bounds.

Table 3
Baseline scenario parameters.

Parameter	Unit	Value	Range
Biodiesel reference	Litre	$1.5e + 009$	$[1e + 009, 2e + 009]$
Feedstock cost growth	Dmnl	0.001	$[-0.05, 0.05]$
Other operational cost growth	Dmnl	0.001	$[-0.05, 0.05]$
Capital cost growth	Dmnl	0.001	$[-0.05, 0.05]$
Water cost growth	Dmnl	0.001	$[-0.05, 0.05]$
Time to change perception	Year	10	$[5, 20]$

4.3. Expert opinion

Qualitative validation using expert opinion was also used in this study to assess the model usefulness, importance and quality. The *BIOTSA* model baseline results and scenarios were presented in a number of lectures and discussions. During these discussions and lectures, a number of suggestions were highlighted for improving the model from both technical and practical points of view.

5. *BIOTSA* model limitations and challenges

Computer simulations only provide a simple representation of the reality that is being investigated. Thus, they do not capture all the inherent aspects of the reality; hence resulting in some limitations. The need to provide a specific focus and data unavailability due to non-existence of biodiesel market in South Africa are the main reasons for the limitations of the *BIOTSA* model. Some of the specific limitations of the *BIOTSA* model are:

5.1. Biodiesel market

The biodiesel production chain consists of crop production, biodiesel production and biodiesel market as illustrated in Fig. 19. The biodiesel market can either be export-oriented or for local consumption. By-products from biodiesel production can be used as animal feed and in the chemical industry. While the *BIOTSA* model considers a project aimed for the export market, the boundary is limited to the crop production and biodiesel production chain (see Fig. 19).

The dynamics of the export market was not taken into account although it could influence the selected sustainability indicators such as air emissions. However, the biodiesel production is not mainly dependent on the outside (export) market but on local conditions such as land availability, community perception and local incentives for investments. On the other hand, a project on large-scale biodiesel production for use at the local market in South Africa would have been an interesting one to investigate. The selection of a project for analysis was however based on the most likely or potential large-scale biodiesel production development in South Africa. Due to the non-existence of large-scale biodiesel production and a market in South, the local market was not considered. This could be attributed to the policy issues and lack of enabling framework to support such a market. It is thus important for future studies to consider the full value chain when there is an enabling framework to support large-scale biodiesel production and market in South Africa.

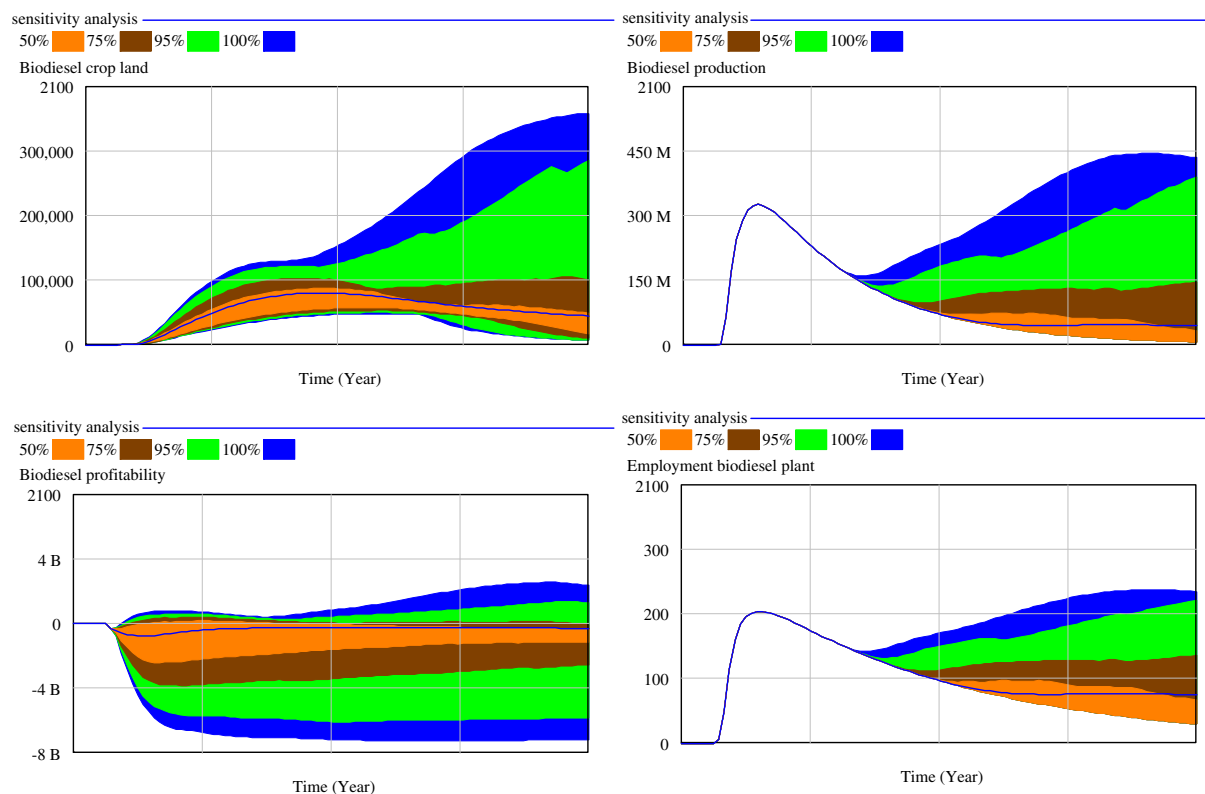


Fig. 18. Sensitivity analysis of cost growth rates result.

5.2. Implicit farming activities

The BIOTSA model assumes that the community can easily alter the fallow land to biofuel crop land as long as they have the acceptance to convert this land. The whole social process such as training of the community on the new farming activity was not considered. This was due to the scope of the study which excluded such analysis.

5.3. Feedstock logistics

There are a number of feedstock logistics that are involved in the biodiesel production such as: biomass collection, pre-processing, storage and transportation. Unfortunately, such level of detail was not included in the BIOTSA model. This was due to the lack of such information in the South African situation. In addition, given the objectives of the model, this level of detail was deemed

unnecessary. However, all these activities that are involved were aggregated to provide the effect of such activities on feedstock cost depending on whether the feedstock would be sourced locally or from importation.

5.4. Employment

Biodiesel production development is also claimed to create employment in the farming communities. Unfortunately, the BIOTSA model does not consider this type of employment and only employment in the biodiesel production processes is considered. This was because the level of information on employment in the whole supply chain is unavailable. In addition, given the land that is planned for biodiesel crop production is in the rural areas, it would be difficult to determine the extent to which this employment would be created and thus this was excluded from the model. However, it is possible and important to include when

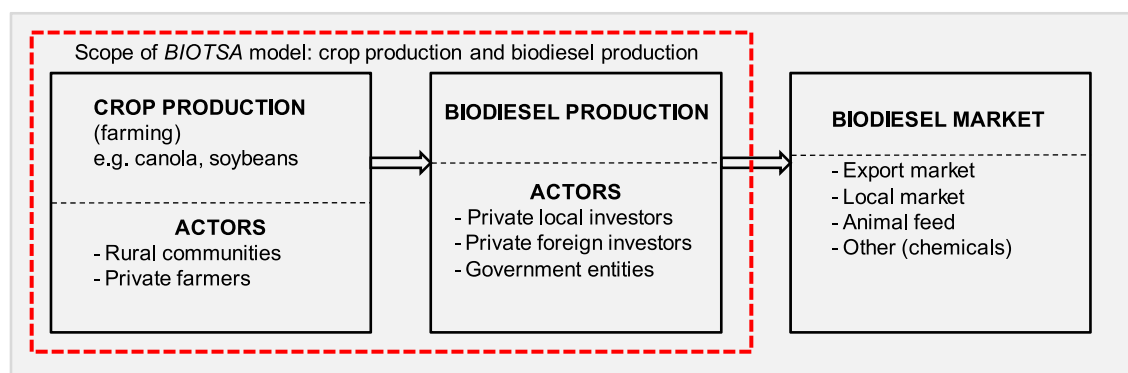


Fig. 19. Illustration of biodiesel production chain.

assessing a medium-scale and small-scale biodiesel production development.

Regardless of the limitations and the challenges discussed, the *BIOTSA* model does provide insights for assessing the impact of biodiesel development on the sustainability indicators and opportunities to improve the value chain, which is summarized in Table 4.

Table 4

Summary of value chain insights from *BIOTSA* model.

Crop production	Biodiesel production
• The need to improve community perception of biodiesel crops benefits, which result from fear and previous bad experiences. • Promoting local feedstock production. • Focussing on non-food land for biodiesel crop production	• Local job creation at biodiesel plant level. • Using by-products as part of income generation outputs. • Government support in the biodiesel production. • Reducing feedstock costs by sourcing it locally.

6. Conclusions

Large-scale biodiesel production in South Africa is non-existent and many studies make future predictions other than the case of reporting historical data. Thus, models become useful in exploring changes in expected energy technology development trends. This paper thus introduced the Bioenergy Technology Sustainability Assessment (*BIOTSA*) model that was developed, based on a system dynamics approach. The *BIOTSA* model was built with the purpose to assess the effect of proposed biodiesel production development in the Eastern Cape on selected sustainability indicators. The *BIOTSA* model was exposed to a number of validation and verification tests namely: structural, behavioural and expert opinion. Generally,

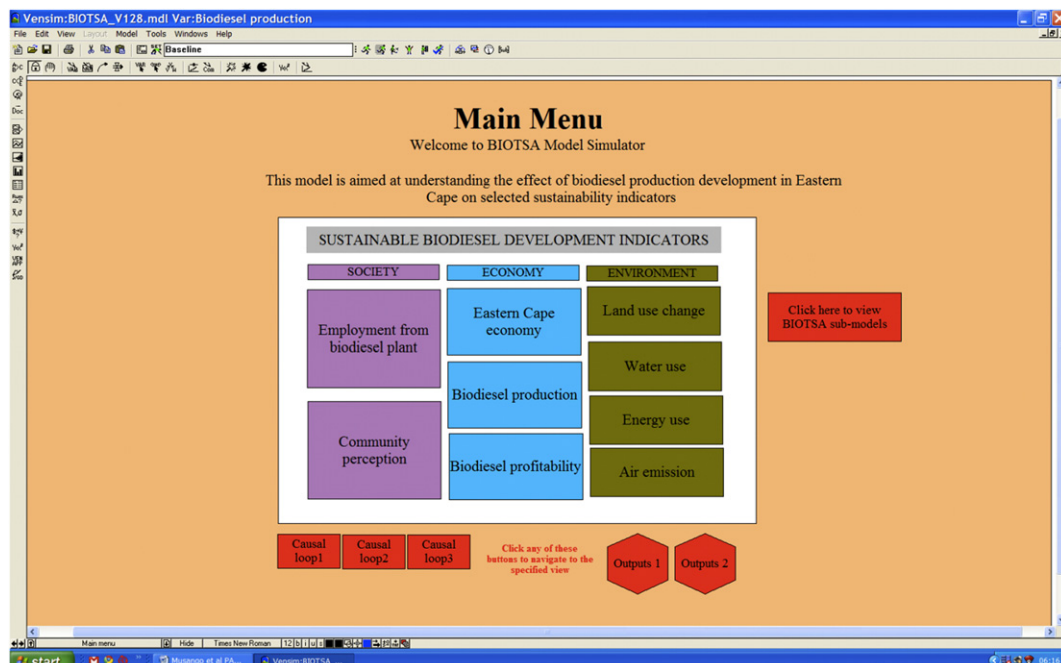
based on these validation tests to which the *BIOTSA* model was exposed, the confidence of the model was improved and the model is thus deemed capable of generating the “right behaviour for the right reasons”.

The *BIOTSA* model does not capture all the inherent aspects of the biodiesel production development and this resulted in some limitation of the model which was discussed in the paper. The reason for not incorporating all these aspects was due to the need to provide a specific focus and data unavailability due to non-existence of biodiesel market in South Africa. While the limitations and the challenges of the *BIOTSA* model are highlighted, the model does provide insights to technology assessment practitioners, technology developers, government agencies and policy makers on the impact of biodiesel development on sustainability indicators. It also provides opportunities for future research. For instance, the *BIOTSA* model can be customized to explore the biodiesel production for local consumption in South Africa as a whole where the biodiesel development will be used locally. In this way, it should provide an investigative tool to assess the impacts of biodiesel production development on sustainability indicators in South Africa. The usability of the *BIOTSA* model is currently being tested with decision-/policy-makers using the VENSIM[®] software interactive graphical user interface as shown Appendix 1.

Acknowledgements

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Appendix 1. *BIOTSA* graphical user interface.



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